# The Effect of 2020 COVID-19 Mortality Rates on 2021 Housing Prices Growth Rate in the US at the County-Level

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## Introduction

The COVID-19 pandemic of 2020 heavily impacted the global economy in a number of ways, and while there is much discussion of the impacts this pandemic had on businesses, inflation on consumer goods, and employment rates, there is less said about the impact it had on the housing market. Accordingly, this paper will seek to answer the question: what effect did county-level per-capita COVID-19 mortality rates in 2020 have on housing prices in 2021 in the United States? This question is particularly interesting and relevant to understanding the economic impacts of the COVID-19 pandemic, because housing prices play a key role in the US economy and heavily impact the lives of US citizens. Housing prices are constantly subject to change based on a number of macroeconomic factors, but this paper will seek to investigate the specific effect 2020 COVID-19 mortality rates had on housing prices, using the 2021 growth rate of the House Price Index (HPI).

## Data Description

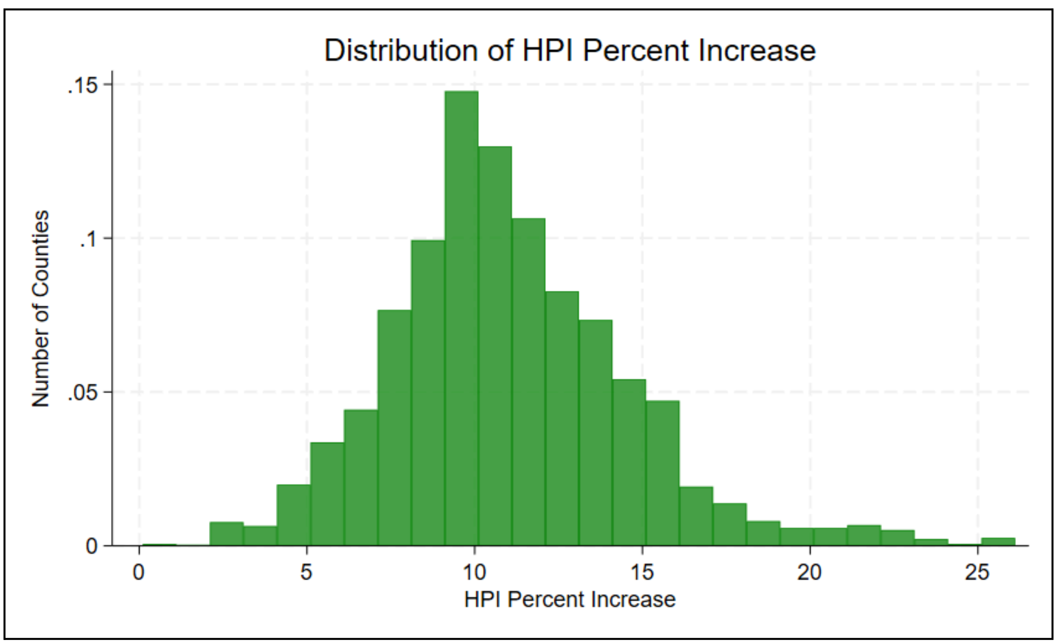
To answer this question, I collected county-level COVID-19 cases and deaths data from the NY Times, county-level population data from the 2020 US Census, ZIP-level HPI data from the Federal Finance Housing Agency, county landmass data from Brian Dill on Kaggle, county-level data on US hospitals in 2020 from the Homeland Infrastructure Foundation-Level Data, and a ZIP-county crosswalk from the US Department of Housing and Urban Development's Office of Policy Development and Research. All of this data was cleaned and merged into one master dataset to run a series of data analyses to answer the research question of this paper.

To construct the outcome variable for this analysis, I used the percent change in county-level HPI from 2020 to 2021. To be specific, I calculated this by using the difference between HPI in 2021 and 2020, dividing it by the 2020 level, and multiplying it by 100 so that the measure is expressed in percentage points. Using percent change rather than just the difference allows for a more meaningful comparison across counties, since housing prices vary substantially across the United States. This approach captures the relative impact of COVID-19 mortality on housing price growth.

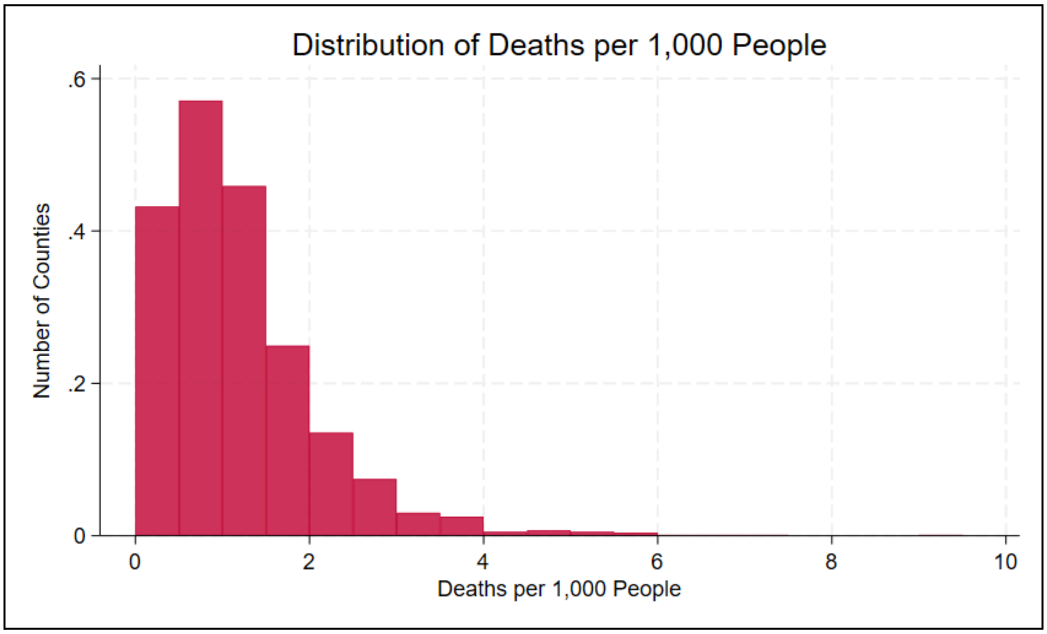
The main independent variable of interest I use is COVID-19 deaths per 1,000 people. I constructed this by dividing total county-level deaths by population and multiplying by 1,000. For control variables, I include COVID-19 cases per 1,000 people, population density, and the number of hospitals at the county level. Cases per 1,000 people are calculated using the same method as deaths. I constructed population density by dividing the population by land area in square miles, and the number of hospitals is computed by summing hospitals within each county using FIPS codes. The following table summarizes the key metrics for these variables.

| Variable | Observations | Mean | Standard Deviation | Minimum | Maximum |
|---|---|---|---|---|---|
| HPI Percent Increase | 3120 | 10.9879 | 3.5877 | 0.1071 | 25.7036 |
| Deaths per 1,000 People | 3120 | 1.1759 | 0.8919 | 0 | 9.0361 |
| Cases per 1,000 People | 3120 | 67.1416 | 27.8459 | 1.9919 | 277.8809 |
| Population Density | 3117 | 219.1566 | 820.5223 | 0.0366 | 18816.66 |
| Number of Hospitals | 3120 | 2.3635 | 5.3078 | 0 | 129 |

This table highlights we have data for about 3,120 counties in our dataset, with the only exception being 3 counties where we don't have the population density. However, these 3,120 counties do not include every county included in the U.S. territories, as some counties were not accounted for in some of my raw datasets. Regardless, we have a sufficiently large sample size to answer the given research question. In this table, it is also interesting to note that not every county had a COVID-19 death and not every county has a hospital. With this in mind, it will be interesting to see what our regression results indicate. However, before diving into any regressions, we will first examine the distribution of HPI percentage increase and deaths per 1,000 people to get a better idea of the key variables we are concerned about.




From the HPI percent increase distribution table, we can see that the distribution appears to mimic a standard normal distribution. Additionally, most counties experienced between a 7 to 16% increase in their HPI, with the rest of the distribution appearing to thin out on both ends of the graph. In the next graph, the Deaths per 1,000 people distribution is far more skewed to the left side of the graph, implying that very few counties experienced high death rates. Thus, the impact that 2020 COVID-19 death rates had on the 2021 HPI percent increase is not blatantly obvious, meaning we will have to use empirical analysis to understand how these death rates affected HPI inflation.




## Empirical Analysis

In order to measure the causal relationship that 2020 COVID-19 mortality rates had on 2021 HPI percent increases on the county-level, I will run a pair of regressions to estimate what impact one death per 1,000 people has on the percent increase of HPI within a county. The first regression I will run will be a simple bivariate regression of HPI percentage increase on COVID-19 deaths per 1,000 people. The model of this regression is specified as follows, where $hpipcinc_i$ is defined as the HPI percent increase from 2020 to 2021, $dp1000_i$ is defined as the COVID-19 deaths per 1,000 people, and $\varepsilon_i$ is the unobserved factors for the ith county in the dataset.

$$hpipcinc_i = \beta_0 + \beta_1 dp1000_i + \varepsilon_i$$

This bivariate regression yields the following results.

| Regressor | Coefficient | Standard Error | Test Statistic | P-Value |
|---|---|---|---|---|
| dp1000 | -0.9594042 | 0.069962 | -13.71 | 0.000 *** |
| Constant (β_0) | 12.11612 | 0.1032497 | 117.35 | 0.000 |

Note: *** indicates that the results are statistically significant at every significance level.

These results suggest that each additional COVID-19 death per 1,000 people in 2020 is associated with a 0.95 percentage point reduction in the growth rate of county-level housing prices. This estimate is very statistically significant, with a test statistic of -13.71 and a p-value of 0.000, however there are likely many other factors within $\varepsilon_i$ that are causing bias within this estimate. To reduce this bias, I will run the multiple regression specified below, where $hpipcinc_i$ $dp1000_i$, and $\varepsilon_i$ are defined as before, while $cp1000_i$ is defined as the COVID-19 cases per 1,000 people, $popdensity_i$ is defined as the number of people per square mile of land, and $numhospitals_i$ is defined as the number of hospitals for the ith county in the dataset.

$$hpipcinc_i = \beta_0 + \beta_1 dp1000_i + \beta_2 cp1000_i + \beta_3 popdensity_i + \beta_4 numhospitals_i + \varepsilon_i$$

I chose to include these three additional variables as controls in my regression since they are all likely correlated with the $hpipcinc_i$ variable. If 2020 COVID-19 death rates impact housing prices in 2021, then 2020 COVID-19 cases rates must also have some impact on these housing prices. Furthermore, higher population densities in counties must impact the demand for housing and thus affect the growth rate of 2021 house prices. Similarly, counties with more hospitals likely impact demand for housing, which also likely affects the growth rate of 2021 house prices. To empirically verify these correlations, the correlation coefficients of $hpipcinc_i$ and $cp1000_i$ is -0.2293; that of $hpipcinc_i$ and $popdensity_i$ is -0.0444 ; and that of $hpipcinc_i$ and $numhospitals_i$ is 0.0501. While these coefficients are not very large, it is important to include them in our regression to limit as much omitted variable bias as possible.

In addition to their correlation with $hpipcinc_i$, it is important that these variables are also correlated with our primary regressor of interest: $dp1000_i$. Of course, COVID-19 cases must be correlated with COVID-19 deaths since it is impossible to die of COVID-19 without developing a case. However, someone who contracts COVID-19 will not necessarily die of COVID-19. Thus, perfect collinearity is not an issue when including $cp1000_i$ in the regression. Next, population density was known to be correlated with COVID-19 deaths in 2020, which is why social distancing was a social health policy implemented throughout the country. Finally, it makes sense for counties with more hospitals to experience fewer COVID-19 deaths due to increased medical access. To empirically verify these correlations, the correlation coefficients of $dp1000_i$ and $cp1000_i$ is 0.5119, that of $dp1000_i$ and $popdensity_i$ is -0.0428, and that of $dp1000_i$ and $numhospitals_i$ is -0.0483. Once again, most of these coefficients are not very large, but it is important to include them in our regression to limit as much omitted variable bias as possible. With this in mind, this multiple regression yields the following results.

| Regressor | Coefficient | Standard Error | Test Statistic | P-Value |
|---|---|---|---|---|
| dp1000 | -0.6467359 | 0.0804244 | -8.04 | 0.000 *** |
| cp1000 | -0.0200558 | 0.0025884 | -7.75 | 0.000 *** |
| popdensity | -0.0004274 | 0.0000825 | -5.18 | 0.000 *** |
| numhospitals | 0.0502411 | 0.0127029 | 3.96 | 0.000 *** |
| Constant (β_0) | 13.07324 | 0.1659956 | 78.76 | 0.000 |

Note: *** indicates that the results are statistically significant at every significance level.

These new results suggest that each additional COVID-19 death per 1,000 people in 2020 is associated with a 0.64 percentage point reduction in the growth rate of county-level housing prices. This estimate is also statistically significant, with a test statistic of -8.04 and a p-value of 0.000. Thus, our new regression still indicates that COVID-19 mortality rates decrease the growth rate of county-level housing prices, but this estimated decrease is not as large in magnitude after removing the omitted variable bias caused by the new variables added to our regression. It is also interesting to note that these additional variables also had a statistically significant impact on the growth rate of housing prices at every significance level, but these impacts are far smaller in magnitude compared to the effect that COVID-19 mortality rates had. Thus, out of the four independent variables in this multiple regression, $dp1000_i$ had the biggest impact on $hpipcinc_i$.

## Conclusion

Based on this data analysis, there is reason to conclude that county-level per-capita COVID-19 mortality rates in 2020 caused a small, yet significant decrease in the HPI growth rate from 2020 to 2021. Of all the variables I chose to include in the multiple regression, the mortality rate had the largest impact on the growth rate compared to the other regressors included in the regression. However, these results do not necessarily measure the true causal relationship these variables had on the 2021 HPI growth rate. There are likely many other factors that affected both the COVID-19 mortality rate and the HPI growth rate that are not included in this refined regression. These omitted variables cause bias that is not accounted for until they are included in our regression. To improve the accuracy of these estimates, more county- or state-level control variables could be added to the regression. Other macro-level variables that change over time could be added to the regression as well, since we are working with data that changes over time. Nonetheless, the data analysis of this paper provides a good starting point for measuring the causal relationship between county-level per-capita 2020 COVID-19 mortality rates and HPI growth rate from 2020 to 2021.